

8086 CPU Architecture & Internal Registers

Q. Explain the internal architecture of the 8086 microprocessor. Describe the functions of the Bus Interface Unit (BIU) and the Execution Unit (EU).

****Definition to Remember****

The ****BIU**** is an asynchronous unit that handles address and data transfers (decodes instructions).

1. Bus Interface Unit (BIU)

The BIU handles all data and address transfers on the system buses.

****Function of BIU****

2. Execution Unit (EU)

The EU is responsible for decoding and executing the fetched instructions.

****Function of EU****
The EU executes the instructions and produces the results.

3. Parallel Operation

Because the BIU and EU operate independently, the BIU continuously fills the 6-byte queue from memory while the EU executes the previously fetched instructions. This prevents the EU from sitting idle waiting for memory fetches, drastically reducing execution time.

****Must-Write Points (for 10 marks)****

1. 8086 Architecture

****Quick Recall****

8086 Architecture

